

Simulations of Gambler's Ruin

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1 Introduction

Arbitrage betting is a strategy by which a player bets on both sides of the same game. This is done by taking advantage of differences in the probabilities offered by competing bookmakers. In theory, provided the bets are placed correctly, this would mean that it is impossible to ever lose a bet. However, oftentimes theory does not match reality and factors such as human error and mistakes made by the bookmakers' website make losses possible. Moreover, there is a learning curve to understand all the terminology, and it takes time to find arbitrage opportunities without the aid of software.

2 Odds And Implied Probability

Understanding the strategy of arbitrage betting begins with knowing how odds and implied probabilities work. There are various ways to write odds, but we will go over two of them, decimal (European odds) and moneyline (American odds). In order to figure out the payout using decimal odds, multiply the odds $\times$ stake = profit. For example, I stake \$100 on team A with odds of 1.8, then I would win \$80. $1.8 \times 100 = \$80$. Moneyline odds are divided into two scenarios. The negative odds are how much you have to stake to win \$100 and the positive odds are how much you earn if you stake \$100. So, in the case of +100 odds you win \$100 for staking \$100. Then for -10000 odds you have to stake \$10,000 to win \$100. The implied probability can be calculated from the bookmakers' odds. For decimal odds, it follows the equation: $P = (\frac{1}{\text{decimal odds}}) \times 100$. For the moneyline odds: $P = (\frac{\text{stake}}{\text{stake} + \text{profit}}) \times 100$.

Implied Probability	Decimal (European)	Moneyline (American)
99.01%	1.01	-10000
80.00%	1.25	-400
75.00%	1.33	-300
66.67%	1.50	-200
55.56%	1.80	-125
50.00%	2.00	+100
45.45%	2.20	+120
40.00%	2.50	+150
33.33%	3.00	+200
13.33%	7.50	+650
5.00%	20.00	+1900

Table 1: Odds & Implied Probabilities

3 Arbitrage Opportunities

An arbitrage opportunity is available when the implied probabilities sum to less than 1. If the implied probabilities sum to 1, you break even. And if they sum less than 1, then you lose money. Let us suppose that FC Barcelona and Real Madrid CF are playing against each other and bookmaker A has the odds as 1.6 for Barcelona and 2.3 for Madrid. Meanwhile, bookmaker B for the same game and the odds as 1.9 for Barcelona and 2.0 for Madrid. The best strategy we can create given these two bookmakers is Barcelona with bookmaker B and Madrid with bookmaker A. To calculate how much to stake on each, we first convert the odds into implied probabilities. So, $\frac{1}{2.3} = 0.4347$, $\frac{1}{1.9} = 0.5263$, Then, we sum the probabilities of 0.5263 and 0.4347. $0.5263 + 0.4347 = 0.961$. We need to normalize this probability so that it is equal to 1 otherwise it is not valid given we are betting on all the possible outcomes of this game. We are assuming ties are not possible in this case. Our total probability is divided with 1 in the numerator and itself in the denominator. $\frac{1}{0.961} = 1.04058$. Then, multiply the previous equation by 1.04058. So, $1.04058 \times (0.5263 + 0.4347) = 0.961 \times 1.04058$. We get $0.54767 + 0.45242 = 1$. Choose our total stake amount, say \$100. Multiply each the total stake \$100 by each respective probability. So, $100 \times 0.54767 = \$54.76$ and $0.45242 \times 100 = \$45.24$. Finally, multiply the odds by each stake

amount. So, $1.9 \times 54.76 = \$104.05$ and $2.3 \times 45.24 = \$104.05$. This gives us a profit of \$4.05 or a return on investment of 4.05%.

	Odds	Stake	Profit
Bookmaker A			
Madrid	1.9	\$54.76	\$104.05
Bookmaker B			
Barcelona	2.3	\$45.24	\$104.05

Table 2: Example Arbitrage Bet

4 Simulator

I have created an arbitrage simulator that more closely models real life by accounting for human error. There are 8 variables that can be adjusted: `starting_bankroll`, `average_stake`, `stake_std`, `average_ROI`, `execution_probability`, `failure_loss_fraction`, `number_of_rounds`, `number_of_simulations`. `Starting_bankroll` is defined as the starting amount of money. `Average_stake` is the average total stake on both sides of the arbitrage. `Stake_std` is the standard deviation of each total stake. The `average_ROI` is the percentage of profit made on each stake. `Execution_probability` is the success rate of a properly executed arbitrage. `Failure_loss_fraction` is the proportion of the stake lost due to an incorrectly placed arbitrage. The `number_of_rounds` is how many times the process is repeated. The `number_of_simulations` is how many individual simulations we computed.

4.1 Results

Three different scenarios were considered. Novice: `Starting_bankroll` = 3000, `average_stake` = 30, `stake_std` = 5, `average_ROI` = 0.05, and `execution_probability` = 0.8. Intermediate: `Starting_bankroll` = 3000, `average_stake` = 50, `stake_std` = 5, `average_ROI` = 0.05, and `execution_probability` = 0.9. Advanced: `Starting_bankroll` = 3000, `average_stake` = 100, `stake_std` = 15, `average_ROI` = 0.05, and `execution_probability` = 0.95.

The results of the three different scenarios evaluated can be summarized using expected value (EV) calculations. $E(X) = \sum_i x_i p_i$. The exe-

cution_probability can be defined as p , while $(1 - p)$ is its compliment, representing the probability of execution failure. A successfully executed stake generates a 5.00% profit from the average_stake and a failure losses 50% of the average_stake. The expected value of any scenario is the following:

$$E(Skill) = p \times (\text{average_ROI} \times \text{average_stake}) \\ + (1 - p) \times (\text{average_stake} \times \text{failure_loss_fraction})$$

$$E(Novice) = 0.8 \times (0.05 \times 30) - (1 - 0.8) \times (30 \times 0.5) = -1.8$$

$$E(Intermediate) = 0.9 \times (0.05 \times 50) - (1 - 0.9) \times (50 \times 0.5) = -0.25$$

$$E(Advanced) = 0.95 \times (0.05 \times 100) - (1 - 0.95) \times (100 \times 0.5) = 2.25$$

Having calculated the expected value the results become obvious and make it clear that without achieving a positive EV making a profit is impossible. An interesting observation to note is that despite the novice having a high failure rate and placing nearly 2,000 bets they still did not lose all of their bankroll. Provided one starts with a big enough bankroll and appropriately proportions their stake size it is possible to leave ample time for improvement. For instance, depending on the improvement rate of a player it might be reasonable to assume starting from a bankroll of \$3000 that they can transition from novice to intermediate and then advanced. If achieved, then arbitrage betting would become a winning strategy.

4.2 Final Thoughts

There are a few improvements that can be made to this simulator. For instance, turning the variable failure_loss_fraction into a distribution of outcomes each with different risk parameters. This could mean taking too long to place a bet on one side, leaving you exposed. Another scenario might be one bookmaker voiding your bet while the other did not.

5 Appendix

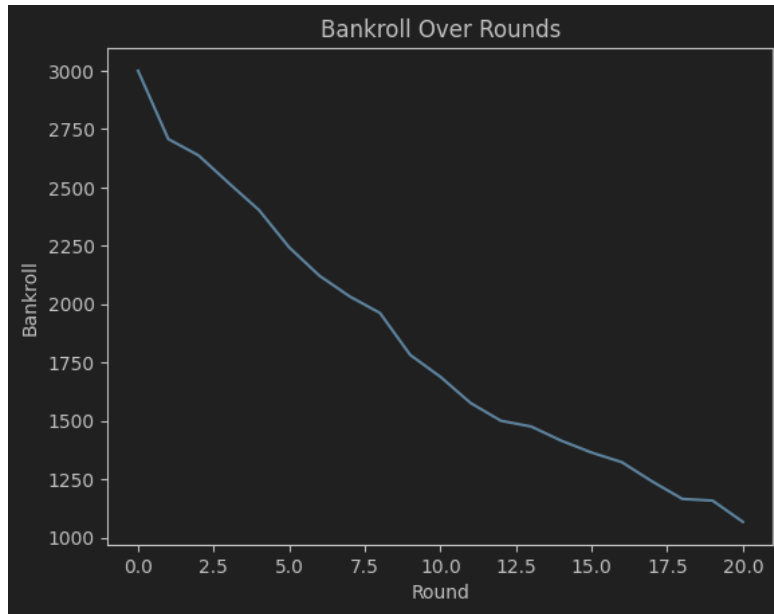


Figure 1: Novice 1 Simulation

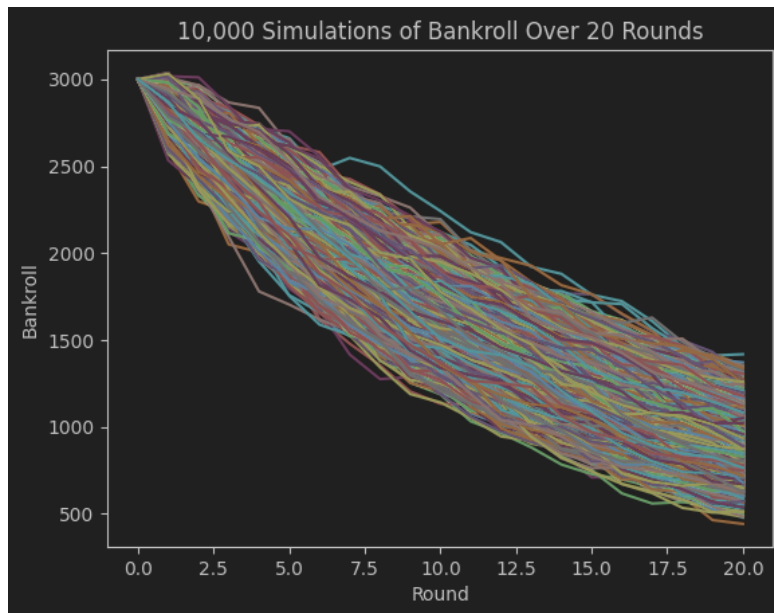


Figure 2: Novice 10,000 Simulations

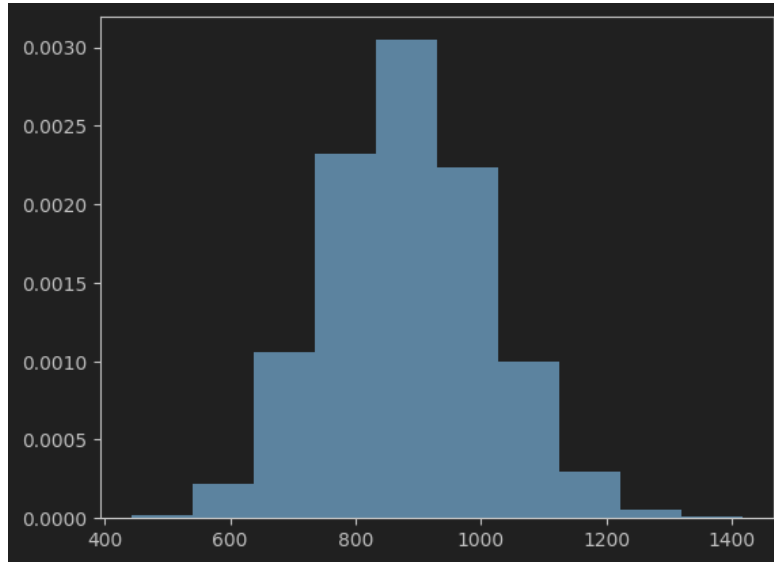


Figure 3: Novice Distribution of 10,000 Simulations

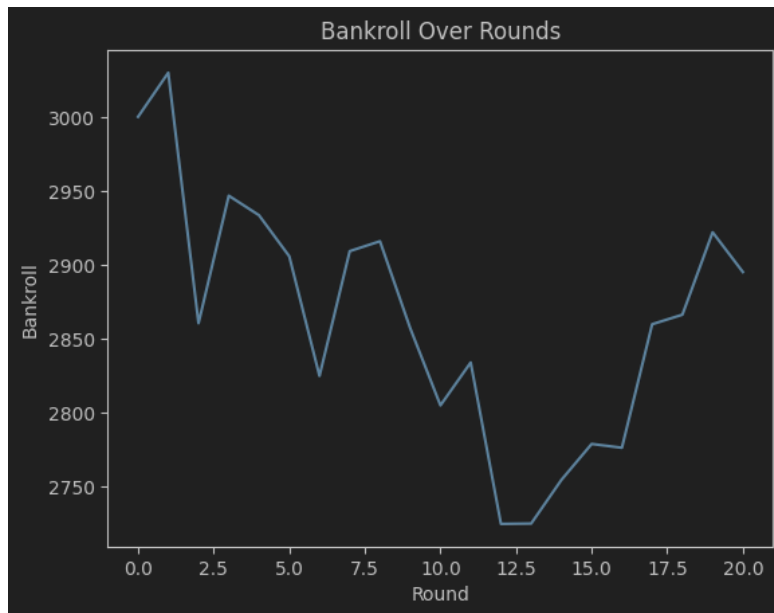


Figure 4: Intermediate 1 Simulation

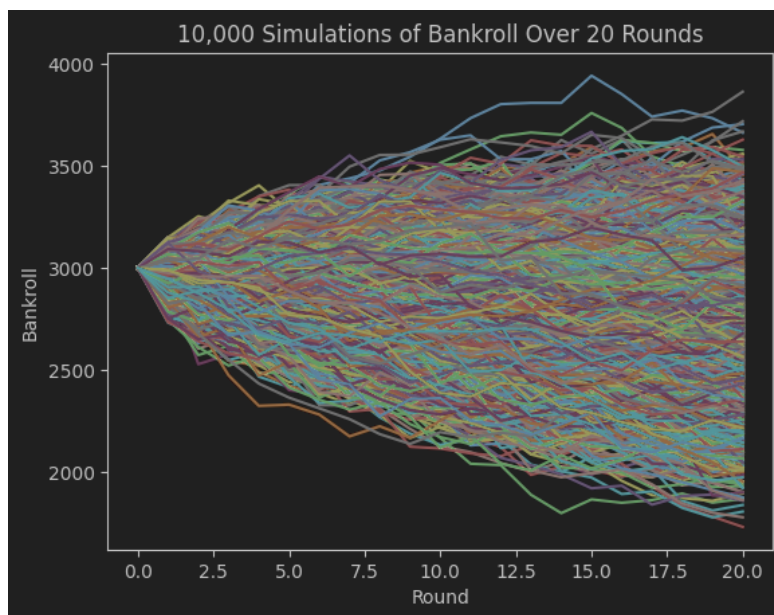


Figure 5: Intermediate 10,000 Simulations

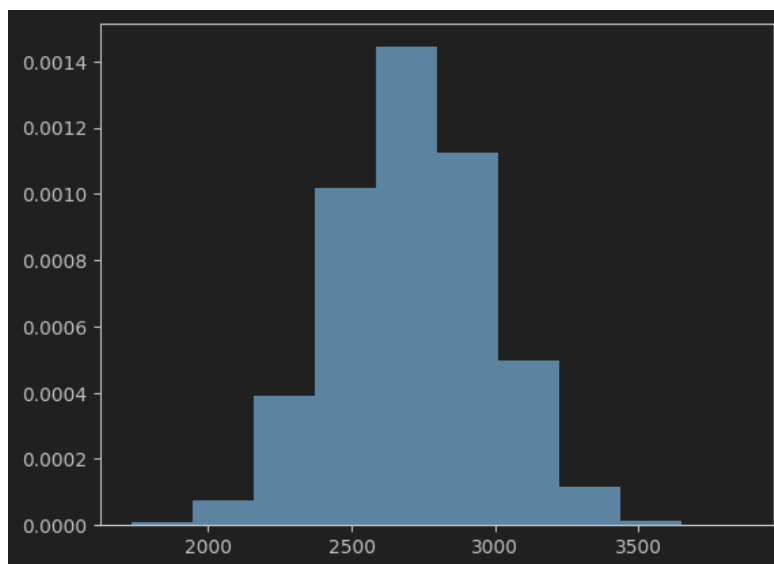


Figure 6: Intermediate Distribution of 10,000 Simulations

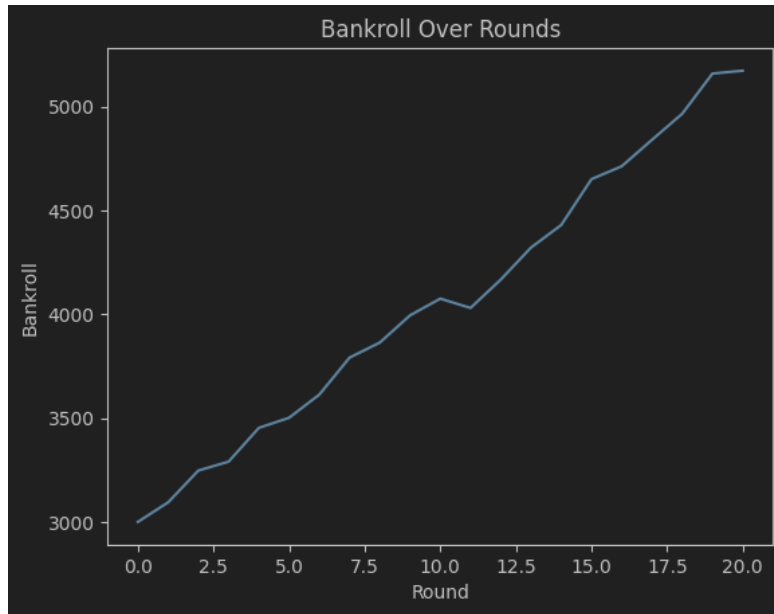


Figure 7: Advanced 1 Simulation

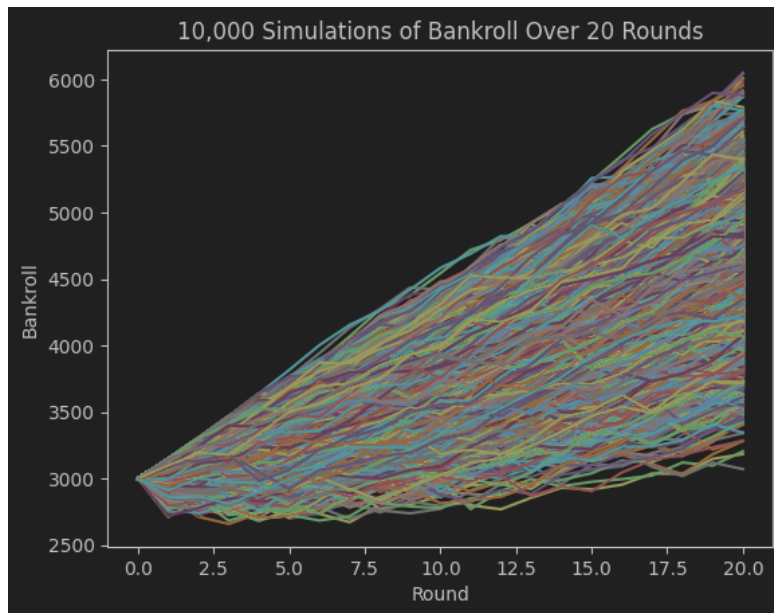


Figure 8: Advanced 10,000 Simulations

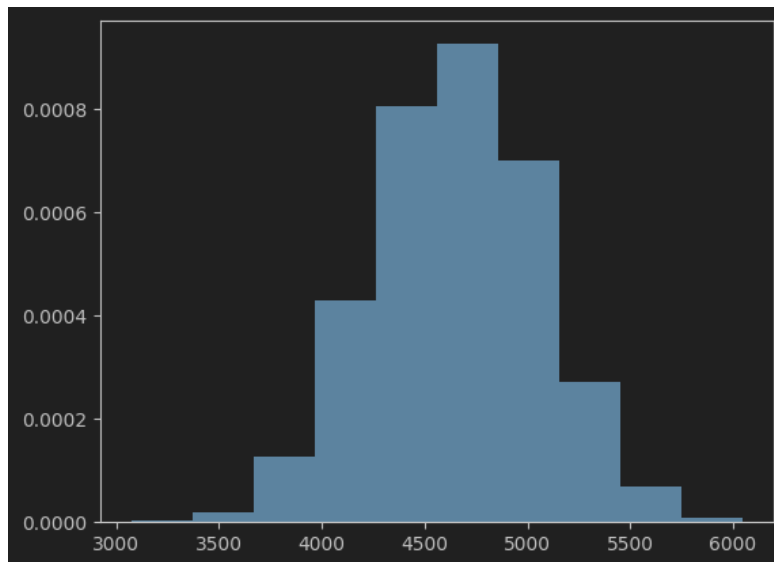


Figure 9: Advanced Distribution of 10,000 Simulations